

Utrecht Summer School  
*Introduction to Complex Systems*

Tuesday projects  
**Simulating patterns: Cellular automata**

## 1 Patterns & panic (*Python*)

Many spatial patterns observed in complex systems can be explained via evolution rules for each individual element of the system. For example, vegetation patterns in ecosystems can be related to plant interactions; cloud formation can be traced to the evolution of individual droplets, and also social segregation can be viewed in this way.

Pattern formation can occur if the complex system has a spatial scale-dependent feedback mechanism: an element's feature is beneficial for other elements over a short range but detrimental over a longer range. This is also known as 'short-range facilitation' or 'short-range activation' and 'long-range inhibition'. As an example, plants can facilitate further plants growing near them by accumulating moisture and nutrients, while 'stealing' water from further regions where plant growth is then inhibited.

One way to model such spatial interactions is via cellular automata (CA). In this project, we will use a CA simulator that implements a given evolution rule for the individual elements.

### 1.1 Emergent patterns

Consider a lattice of  $M \times N$  cells with periodic boundaries (i.e. the right-side of the domain is connected to the left-side and the bottom-side is connected to the top-side; this is a 'trick' often used to overcome the unwanted effects of having fixed boundaries). All cells can be either passive (0) or active (1). With  $A_{ij}(t)$  we denote the value (either 0 or 1) of the site at row  $i$ , column  $j$  at time  $t$ . Now, the evolution rule is based on the spatial scale-dependent feedback. For each cell, we compute the experienced total facilitation/inhibition  $s_{ij}$ , where every active state within activation range  $R_a$  contributes  $+w_a$  and every active state within inhibition range  $R_i$  contributes  $-w_i$ . If  $s > 0$ , the cell will be active at the next time step; otherwise it will be passive. Mathematically, this evolution rule is given by

$$A_{ij}(t+1) = \begin{cases} 1, & s_{ij}(t) > 0; \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

where  $s$  is given by

$$s_{ij}(t) = \sum_{(k,l) \in N_a(i,j)} A_{kl}(t)w_a - \sum_{(k,l) \in N_i(i,j)} A_{kl}(t)w_i, \quad (2)$$

and  $N_a(i, j)$  respectively  $N_i(i, j)$  are the collection of cells within activation respectively inhibition range:

$$N_a(i, j) = \{(k, l) : \text{the distance between } (i, j) \text{ and } (k, l) \text{ is smaller than } R_a\}; \quad (3)$$

$$N_i(i, j) = \{(k, l) : \text{the distance between } (i, j) \text{ and } (k, l) \text{ is larger than } R_a \text{ and smaller than } R_i\}. \quad (4)$$

To simulate this system, we prepared code for you that provides a `CellularAutomaton` model. Use the Jupyter notebook `pattern_formation.ipynb` to get started. You do not need to modify the source code in `cellular_automaton.py` but you are of course welcome to.

### 1.1.1 The spatial scale-dependent feedback

The above description depends on the parameters  $R_a$ ,  $R_i$ ,  $w_a$  and  $w_i$ . What kind of conditions do you need to impose on those to allow for pattern formation? Make a diagram for the scale-dependent feedback; i.e. the influence of an ‘on’ state cell on  $s_{ij}(t)$  depending on its distance to the cell  $(i, j)$ . In the code, the system is modelled on a lattice with periodic boundaries. Does this influence the results? Think about when this is appropriate and when this is not?

### 1.1.2 Equilibrium states

Run the model for some parameter values that lead to patterns. Will the system converge to an equilibrium state? How does it look like?

### 1.1.3 Dependence on initial conditions

Determine the dependence on the precise initial configuration. For instance, do different patterns emerge when you re-run the code with the same parameter values? Do different patterns emerge for different values of  $p$ ? (Recall that  $p$  denotes the approximate fraction of the initial state that is in the ‘on’ state.)

### 1.1.4 Parameter sensitivity

Experiment with different parameter values and vary the values of  $R_a$ ,  $R_i$ ,  $w_a$  and  $w_i$ . How do the emergent patterns change with these values?

### 1.1.5 Additional suggestions

If time permits, try to come up with a way to quantify patterns and compare the emergent patterns quantitatively.

## 1.2 Panic

We now turn our attention to another phenomenon that can be modelled using cellular automata. Let us say that every cell in the lattice now represents a person who can be either panicked (the ‘active’ state) or calm (the ‘passive’ state). By stipulating some rule to determine when a person becomes calm, we can now consider whether and how panic spreads through the system.

We consider the ‘droplet rule’ for the evolution in this system: for any given person, look at the surrounding individuals (typically taken as the 8 direct neighbouring cells) and count how many of them are panicked. If at least  $q$  people in your surroundings (including you) are panicked (e.g.  $q = 3$  or 4), you become or remain panicked at the next timestep; otherwise you will remain calm.

### 1.2.1 Implementation

Implement the droplet rule in the code. For this you will need to directly modify the function `CA_evolution`.

HINT: Try to think of the pattern formation system from the first part and think about how this rule is different. Are there activators/inhibitors and what is their range? How large does the strength  $s_{ij}(t)$  need to be in order for the person at  $(i, j)$  to become panicked? You can implement this with only one change to the evolution function and changing the parameters of the scale-dependent feedback.

### 1.2.2 Dependence on initial conditions

Run the model for various values of  $p$ . Do you see different behaviour? Why?

### 1.2.3 Quantification of panic spread

Find a way to quantify the end state of the system (i.e. define some order parameter). Compute its value for various values of  $p$ . Does this system have a phase transition? If so, around which  $p$ -value does this happen?

### 1.2.4 Additional suggestions

If time permits, feel free to experiment with other rules. For example, what happens when you change the number of required panicked neighbours from 3 to another number?

Alternatively, you can also look into the behaviour of your order parameter as the value of  $p$  approaches the critical value at which the phase transition happens.

## 2 Forest fires (*Julia, advanced*)

In this problem, we will study a cellular automaton (CA) that models the effect of wildfires on forest and grassland. The model is described by Wuyts and Sieber (2024) (a PDF of the paper is included in the folder).

We consider a square domain with  $N = L \times L$  grid cells. Each grid cell can be in one of four possible states: forest (F), grass (G), burning (B) or ashes (A). The main idea is that fire spreads quickly on grassland but much slower in the forest. Also, a fire is unlikely to ignite inside the forest. This means that the ecosystem gets less affected by fires the more forest it contains, especially if the forest *percolates* across the main such that it encloses grass patches.

Take a look at the paper for more details (see section “The FGBA Probabilistic Cellular Automaton”, Table 1, and Fig. 1A). Use the notebook `forest_fires.ipynb` to get started.

### 2.1 Research questions

1. What are the qualitative dynamics produced by the FGBA model under default parameter settings?
2. What are the different timescales in the system? Make sure you understand the difference between physical time steps and the number of grid cell events (where a cell changes state).
3. How do the dynamics change if you vary the transition rate parameters?
4. Does the system reach an equilibrium over time? How can you quantify this equilibrium?

5. Set the fire ignition rate to  $N\phi = 0.3$  and all other parameters to their default value. Explore the sensitivity to different initial conditions (distributions of F and G). In particular, try initial conditions that differ by their forest area fraction. Do you find multistability (more than one equilibrium depending on initial condition)?
6. If you have time: can you quantify the multistable regime as a function of the fire ignition rate?